

IMU-RGBD Camera Extrinsic Calibration: Observability Analysis and Consistency Improvement

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Abstract—In this paper, we address the problem of extrinsically calibrating an inertial measurement unit (IMU) with respect to an RGBD sensor. In particular, we study the observability of the nonlinear IMU-RGBD calibration system and prove that the calibration parameters are observable given observations to a single point feature. Moreover, we show that the system has four unobservable directions corresponding to the global translation and rotations about the gravity vector. Based on the results of the observability analysis, we design a consistency-improved, observability constrained (OC) extended Kalman filter (EKF)-based estimator for calibrating the sensor pair while at the same time tracking its pose and creating a 3D map of the environment. Finally, we validate the key findings of the observability analysis and assess the performance of the OC-EKF estimator both in simulation and experimentally.

I. INTRODUCTION AND RELATED WORK

In many robotic applications (e.g., search and rescue, environmental monitoring, planetary exploration, etc.), it is necessary to precisely know the robot's 3D position and orientation (pose). Over short periods of time, a robot can track its pose by fusing rotational velocity and linear acceleration measurements provided by an IMU. However, the integration of the bias and noise contaminating the IMU signals, makes the resulting estimates unreliable. For this reason, most inertial navigation systems (INS) rely on GPS measurements for bounding the estimation errors. For robots operating in GPS-denied areas, alternative sensors, such as cameras, have been employed for aiding INS, in what is known as vision-aided (V)-INS. Specifically, by processing observations of point features detected in images, a camera can provide periodic corrections to the inertial estimates (e.g., [1]). Although the small size, low weight and power requirements of cameras make them ideal for mobile robots, their inability to provide the depth to the scene poses certain challenges. In particular, a feature needs to be reliably tracked across multiple images and the baseline between them must be accurately estimated in order to determine the depth to it. Moreover, imprecise depth estimation can cause large linearization errors which can reduce the estimation accuracy and even lead to divergence.

An alternative to V-INS would be to fuse IMU measurements with observations from an RGBD camera, such as the Microsoft Kinect [2], which directly measures the depth to the scene. To do so, however, one needs to precisely

determine the 6 degrees of freedom (dof) transformation between the two sensors. Although numerous algorithms have been developed for IMU-noise characterization (e.g., [3]) and for intrinsically calibrating cameras (e.g., [4]) and RGBD sensors (e.g., [5]), to the best of our knowledge, there exists no algorithm for IMU-RGBD *extrinsic* calibration. Instead, one could potentially use methods designed for IMU-camera calibration (e.g., [6], [7], [8], [9], [10]). However, these would result in sub-optimal calibration accuracy since they would neglect the depth information available in the RGBD data. Furthermore, to date the conditions under which the IMU-RGBD calibration parameters are observable are not known. Previous efforts to study the observability properties of the IMU-camera calibration system have either relied on known calibration targets [8], or have assumed that the camera's roll and pitch angles are perfectly known and at least 3 point features are detected by the camera [9].

In order to address these limitations, this paper makes the following key contributions:

- We study the observability of the nonlinear system describing the IMU-RGBD camera calibration and for the first time prove that although the system is unobservable, the unknown 6-dof transformation lies within its observable subspace and can be determined even when only one (previously unknown) point feature is detected. In this analysis, we do not make any assumptions about the prior location of the feature or the pose of the camera and do not rely on inferred camera measurements as is the case in [8], [9]. This is achieved by employing a new methodology for studying the observability of nonlinear systems [11] which employs appropriate change of variables for isolating the unobservable modes of the system.
- We analytically determine the vectors spanning the unobservable directions of the IMU-RGBD calibration system and use them in order to improve the consistency of the EKF designed for estimating the unknown calibration parameters.
- We validate the performance of the developed estimator, both in simulation and experimentally, when performing simultaneous localization, mapping, and IMU-RGBD calibration.

The rest of the paper is structured as follows. In Section II-A, we briefly describe the method of Hermann and Krener [12] for studying the observability of nonlinear systems and discuss the challenges for proving that a system is unobservable when following this method. In Section II-B, we present

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the approach introduced in [11] that employs change of variables to appropriately factorize the observability matrix and determine its unobservable directions. In Section III, we apply this method to the specific problem of IMU-RGBD calibration and show that although the calibration parameters are observable, the global position and rotation about the gravity vector are not. In Section IV, we present the observability-constrained (OC) EKF algorithm we developed for determining the unknown transformation and briefly discuss how its consistency [13] is improved using the results of Section III. In Section V, we assess the performance of the OC-EKF algorithm both in simulation and experimentally and verify its consistency. Finally, Section VI, presents the conclusions of this work and provides possible directions of future research.

II. NONLINEAR SYSTEM OBSERVABILITY ANALYSIS

In this section, we provide a brief overview of the method in [12] for studying the observability of nonlinear systems and then introduce a new methodology for determining its unobservable directions.

A. Observability Analysis with Lie Derivatives

Consider a nonlinear, continuous-time system:

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{f}_0(\mathbf{x}) + \sum_{i=1}^l \mathbf{f}_i(\mathbf{x})u_i \\ \mathbf{y} = \mathbf{h}(\mathbf{x}) \end{cases} \quad (1)$$

where $\mathbf{u} = [u_1 \dots u_l]^T$ is its control input, $\mathbf{x} = [x_1 \dots x_m]^T$ is the system's state vector, $\mathbf{y}$ is the system output, and $\mathbf{f}_i, i = 0, \dots, l$ are the process functions. The zeroth-order Lie derivative of a measurement function $\mathbf{h}$ is defined as the function itself [12]:

$$\mathcal{L}^0 \mathbf{h} = \mathbf{h}(\mathbf{x}) \quad (2)$$

while the span of the i th order Lie derivative is defined as:

$$\nabla \mathcal{L}^i \mathbf{h} = \begin{bmatrix} \frac{\partial \mathcal{L}^i \mathbf{h}}{\partial x_1} & \frac{\partial \mathcal{L}^i \mathbf{h}}{\partial x_2} & \dots & \frac{\partial \mathcal{L}^i \mathbf{h}}{\partial x_m} \end{bmatrix} \quad (3)$$

For any i th order Lie derivative, $\mathcal{L}^i \mathbf{h}$, the $i+1$ th order Lie derivative $\mathcal{L}_{\mathbf{f}_j}^{i+1} \mathbf{h}$ with respect to a process function $\mathbf{f}_j$ can be computed as:

$$\mathcal{L}_{\mathbf{f}_j}^{i+1} \mathbf{h} = \nabla \mathcal{L}^i \mathbf{h} \cdot \mathbf{f}_j \quad (4)$$

The observability matrix $\mathcal{O}$ of system (1) is defined as a matrix with block rows the span of the Lie derivatives of (1), i.e.,

$$\mathcal{O} = \begin{bmatrix} \nabla \mathcal{L}^0 \mathbf{h} \\ \nabla \mathcal{L}_{\mathbf{f}_1}^1 \mathbf{h} \\ \nabla \mathcal{L}_{\mathbf{f}_2}^2 \mathbf{h} \\ \nabla \mathcal{L}_{\mathbf{f}_3 \mathbf{f}_k}^3 \mathbf{h} \\ \vdots \end{bmatrix} \quad (5)$$

where $i, j, k = 1, \dots, l$. Based on [12], to prove that a system is observable, it suffices to show that any submatrix of $\mathcal{O}$ comprising a subset of its rows is of full column rank. In contrast, to prove that a system is unobservable and find its unobservable directions, we need to: (i) show that the infinitely many block rows of $\mathcal{O}$ can be written as

a linear combination of a subset of its block rows, which form a submatrix $\mathcal{O}'$. (ii) find the nullspace of $\mathcal{O}'$ in order to determine the system's unobservable directions. Although accomplishing (ii) is fairly straightforward, achieving (i) is extremely challenging especially for high-dimensional systems, such as the one describing the IMU-RGBD camera calibration.

To address this issue, in the following section, we present a new methodology that relies on change of variables for proving that a system is unobservable and finding its unobservable directions.

B. Observability Analysis with Basis Functions

We start by proving the following:

Theorem 1: Assume that there exists a nonlinear transformation $\beta(\mathbf{x}) = [\beta_1(\mathbf{x})^T \dots \beta_l(\mathbf{x})^T]^T$ (i.e., a set of basis functions) of the variable $\mathbf{x}$ in (1), such that:

(A1) $\beta_1(\mathbf{x}) = \mathbf{h}(\mathbf{x})$;

(A2) $\frac{\partial \beta}{\partial \mathbf{x}} \cdot \mathbf{f}_i, i = 0, \dots, l$ is a function of β ;

(A3) The system:

$$\begin{cases} \dot{\beta} = \mathbf{g}_0(\beta) + \sum_{i=1}^l \mathbf{g}_i(\beta)u_i \\ \mathbf{y} = \mathbf{h} = \beta_1 \end{cases} \quad (6)$$

where $\mathbf{g}_i(\beta) = \frac{\partial \beta}{\partial \mathbf{x}} \mathbf{f}_i(\mathbf{x}), i = 0, \dots, l$, is observable.

Then:

(i) The observability matrix of (1) can be factorized as: $\mathcal{O} = \Xi \cdot \mathbf{B}$, where Ξ is the observability matrix of system

(6) and $\mathbf{B} \triangleq \frac{\partial \beta}{\partial \mathbf{x}}$

(ii) $null(\mathcal{O}) = null(\mathbf{B})$

Proof: (i) Based on the chain rule, the span of any Lie derivative $\nabla \mathcal{L}^i \mathbf{h}$ can be written as:

$$\nabla \mathcal{L}^i \mathbf{h} = \frac{\partial \mathcal{L}^i \mathbf{h}}{\partial \mathbf{x}} = \frac{\partial \mathcal{L}^i \mathbf{h}}{\partial \beta} \frac{\partial \beta}{\partial \mathbf{x}} \quad (7)$$

Thus the observability matrix $\mathcal{O}$ of (1) can be factorized as:

$$\mathcal{O} = \begin{bmatrix} \nabla \mathcal{L}^0 \mathbf{h} \\ \nabla \mathcal{L}_{\mathbf{f}_1}^1 \mathbf{h} \\ \nabla \mathcal{L}_{\mathbf{f}_2}^2 \mathbf{h} \\ \nabla \mathcal{L}_{\mathbf{f}_3 \mathbf{f}_k}^3 \mathbf{h} \\ \vdots \end{bmatrix} = \begin{bmatrix} \frac{\partial \mathcal{L}^0 \mathbf{h}}{\partial \beta} \\ \frac{\partial \mathcal{L}_{\mathbf{f}_1}^1 \mathbf{h}}{\partial \beta} \\ \frac{\partial \mathcal{L}_{\mathbf{f}_2}^2 \mathbf{h}}{\partial \beta} \\ \frac{\partial \mathcal{L}_{\mathbf{f}_3 \mathbf{f}_k}^3 \mathbf{h}}{\partial \beta} \\ \vdots \end{bmatrix} \frac{\partial \beta}{\partial \mathbf{x}} = \Xi \cdot \mathbf{B} \quad (8)$$

Next we prove that Ξ is the observability matrix of the system (6) by induction.

To distinguish the Lie derivatives of system (1), let $\mathcal{J}$ denote the Lie derivatives of system (6). Then, the span of its zeroth order Lie derivative is:

$$\nabla \mathcal{J}^0 \mathbf{h} = \frac{\partial \mathbf{h}}{\partial \beta} = \frac{\partial \mathcal{L}^0 \mathbf{h}}{\partial \beta} \quad (9)$$

which corresponds to the first block row of Ξ .

Assume that the span of the i th order Lie derivative of (6) along any direction can be written as $\nabla \mathcal{J}^i \mathbf{h} = \frac{\partial \mathcal{L}^i \mathbf{h}}{\partial \beta}$, which corresponds to a block row of Ξ .

Then the span of the $i+1$ th order Lie derivative $\nabla \mathcal{J}_{\mathbf{g}_j}^{i+1} \mathbf{h}$

with respect to the process function $\mathbf{g}_j$ can be computed as:

$$\begin{aligned}\nabla \mathcal{J}_{\mathbf{g}_j}^{i+1} \mathbf{h} &= \frac{\partial \mathcal{J}_{\mathbf{g}_j}^{i+1} \mathbf{h}}{\partial \beta} = \frac{\partial (\nabla \mathcal{J}^i \mathbf{h} \cdot \mathbf{g}_j)}{\partial \beta} = \frac{\partial (\frac{\partial \mathcal{J}^i \mathbf{h}}{\partial \beta} \cdot \frac{\partial \beta}{\partial \mathbf{x}} \mathbf{f}_j(\mathbf{x}))}{\partial \beta} \\ &= \frac{\partial (\frac{\partial \mathcal{J}^i \mathbf{h}}{\partial \mathbf{x}} \cdot \mathbf{f}_j(\mathbf{x}))}{\partial \beta} = \frac{\partial \mathcal{J}_{\mathbf{f}_j}^{i+1} \mathbf{h}}{\partial \beta}\end{aligned}\quad (10)$$

which is also a block row of Ξ . Therefore, we conclude Ξ is a matrix whose rows are the span of all the Lie derivatives of system (6), thus the observability matrix of system (6). ■

Proof: (ii) From $\mathcal{O} = \Xi \mathbf{B}$, we have $\text{null}(\mathcal{O}) = \text{null}(\mathbf{B}) + \text{null}(\Xi) \cap \text{range}(\mathbf{B})$. Moreover, from condition A3 system (6) is observable, and Ξ is of full column rank. Therefore $\text{null}(\mathcal{O}) = \text{null}(\mathbf{B})$. ■

With the basis functions, the unobservable directions can be determined with much less effort. To find a system's unobservable directions, we first need to define the basis functions that satisfy conditions (A1) and (A2), and prove that the infinite-dimensional matrix Ξ has full column rank, which is condition (A3). Once all the conditions are satisfied, the unobservable directions of (1) correspond to the nullspace of matrix $\mathbf{B}$, which has finite dimension and thus it is easy to analyze.

In the following sections, we will leverage *Theorem 1* to prove that the IMU-RGBD camera calibration system is unobservable and find its unobservable directions. To do this, in Section III-A we first show the formulation of the IMU-RGBD camera calibration nonlinear system. In Section III-B, we find the set of basis functions that satisfy conditions (A1) and (A2) of *Theorem 1*, and construct the basis function system as in (6) for this particular problem. In Section III-C, we prove that the observability matrix Ξ for the basis function system is of full column rank, which is condition (A3) in *Theorem 1*. Lastly, we determine the unobservable directions of the IMU-RGBD camera calibration system by finding the nullspace of matrix $\mathbf{B}$.

III. OBSERVABILITY ANALYSIS OF THE IMU-RGBD CAMERA CALIBRATION SYSTEM

In this section, we present the observability analysis for the IMU-RGBD camera calibration problem using basis functions. Since the Lie derivatives for all the landmarks have identical form, for simplicity we keep only one landmark in the state vector. The extension to multiple landmarks is described in [14].

A. System Model

In the IMU-RGBD camera calibration problem, the state vector we estimate is:

$$\mathbf{x} = [{}^I \mathbf{s}_G^T \quad {}^G \mathbf{v}_I^T \quad {}^G \mathbf{p}_I^T \quad {}^G \mathbf{p}_f^T \quad \mathbf{b}_a^T \quad \mathbf{b}_g^T \quad {}^C \mathbf{s}_I^T \quad {}^I \mathbf{p}_C^T]^T$$

where ${}^I \mathbf{s}_G$ is the Cayley-Gibbs-Rodriguez parameterization [15] representing the orientation of the global frame $\{G\}$ in the IMU's frame of reference $\{I\}$, ${}^G \mathbf{v}_I$ and ${}^G \mathbf{p}_I$ represent the velocity and position of $\{I\}$ in $\{G\}$, $\mathbf{b}_a$ and $\mathbf{b}_g$ are the biases in the gyroscope and accelerometer measurements, ${}^C \mathbf{s}_I$ represents the orientation of $\{I\}$ in the RGBD camera's

frame of reference $\{C\}$, ${}^I \mathbf{p}_C$ is the position of $\{C\}$ in $\{I\}$, and ${}^G \mathbf{p}_f$ represents the estimated landmark in $\{G\}$.

1) *System propagation model:* The system model describing the time evolution of the states is:

$$\begin{aligned}{}^I \dot{\mathbf{s}}_G(t) &= \frac{1}{2} \mathbf{D} ({}^I \boldsymbol{\omega}(t) - \mathbf{b}_g(t)) \\ {}^G \dot{\mathbf{v}}_I(t) &= {}^G \mathbf{a}(t) = {}^G \mathbf{g} + \mathbf{C}({}^I \mathbf{s}_G(t))^T ({}^I \mathbf{a}(t) - \mathbf{b}_a(t)) \\ {}^G \dot{\mathbf{p}}_I(t) &= {}^G \mathbf{v}_I(t) \quad {}^G \dot{\mathbf{p}}_f(t) = \mathbf{0}_{3 \times 1} \\ \dot{\mathbf{b}}_a(t) &= \mathbf{0}_{3 \times 1} \quad \dot{\mathbf{b}}_g(t) = \mathbf{0}_{3 \times 1} \\ {}^C \dot{\mathbf{s}}_I(t) &= \mathbf{0}_{4 \times 1} \quad {}^I \dot{\mathbf{p}}_C(t) = \mathbf{0}_{3 \times 1}\end{aligned}\quad (11)$$

where $\mathbf{C}(\mathbf{s})$ represents the corresponding rotation matrix of $\mathbf{s}$, ${}^I \boldsymbol{\omega}(t) = [\omega_1 \quad \omega_2 \quad \omega_3]^T$ and ${}^I \mathbf{a}(t) = [a_1 \quad a_2 \quad a_3]^T$ are the IMU rotational velocity and linear acceleration expressed in $\{I\}$, $\mathbf{b}_g(t)$ and $\mathbf{b}_a(t)$ are the gyroscope and accelerometer biases, $\frac{1}{2} \mathbf{D} \triangleq \frac{\partial \mathbf{s}}{\partial \boldsymbol{\theta}}$, $\mathbf{D} = (\mathbf{I} + [\mathbf{s} \times] + \mathbf{s} \mathbf{s}^T)$, and:

$$\boldsymbol{\Omega}({}^I \boldsymbol{\omega}(t)) = \begin{bmatrix} -[{}^I \boldsymbol{\omega} \times] & {}^I \boldsymbol{\omega} \\ -{}^I \boldsymbol{\omega}^T & 0 \end{bmatrix}, {}^I \boldsymbol{\omega}(t) = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}$$

where $\boldsymbol{\theta} = \alpha \mathbf{k}$ represents a rotation by an angle α around the axis $\mathbf{k}$.

2) *System measurement model:* The RGBD camera can directly measure the 3D position of the landmark ${}^C \mathbf{p}_f$ in its own frame of reference as:

$$\mathbf{z} = {}^C \mathbf{p}_f = \mathbf{C}({}^C \mathbf{s}_I) \mathbf{C}({}^I \mathbf{s}_G) ({}^G \mathbf{p}_f - {}^G \mathbf{p}_I) - \mathbf{C}({}^C \mathbf{s}_I) {}^I \mathbf{p}_C \quad (12)$$

B. Determining the System's Basis Functions

In this section, we define the basis functions for the IMU-RGBD camera calibration system that satisfy conditions (A1), (A2) in *Theorem 1*, and present system (6) with basis functions of this particular problem. In the next section, we will show this system is observable.

For simplicity, we retain only a few of the subscripts and superscripts in the state elements and write the system state vector as:

$$\mathbf{x} = [\mathbf{s}^T \quad \mathbf{v}^T \quad \mathbf{p}^T \quad \mathbf{p}_f^T \quad \mathbf{b}_a^T \quad \mathbf{b}_g^T \quad \mathbf{s}_C^T \quad \mathbf{p}_C^T]^T$$

Then the IMU-RGBD camera calibration system can be rewritten as:

$$\begin{bmatrix} \dot{\mathbf{s}} \\ \dot{\mathbf{v}} \\ \dot{\mathbf{p}} \\ \dot{\mathbf{p}}_f \\ \dot{\mathbf{b}}_a \\ \dot{\mathbf{b}}_g \\ \dot{\mathbf{s}}_C \\ \dot{\mathbf{p}}_C \end{bmatrix} = \underbrace{\begin{bmatrix} -\frac{1}{2} \mathbf{D} \mathbf{b}_g \\ \mathbf{g} - \mathbf{C}^T \mathbf{b}_a \\ \mathbf{v} \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}}_{\mathbf{f}_0} + \underbrace{\begin{bmatrix} \frac{1}{2} \mathbf{D} \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}}_{\mathbf{f}_1} \boldsymbol{\omega} + \underbrace{\begin{bmatrix} \mathbf{0} \\ \mathbf{C}^T \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}}_{\mathbf{f}_2} \mathbf{a}$$

$$\mathbf{z} = \mathbf{C}_C \mathbf{C}(\mathbf{p}_f - \mathbf{p}) - \mathbf{C}_C \mathbf{p}_C \quad (13)$$

where $\mathbf{C} \triangleq \mathbf{C}(\mathbf{s})$, $\mathbf{C}_C \triangleq \mathbf{C}(\mathbf{s}_C)$. Note that $\mathbf{f}_0$ is a 24×1 vector, while $\mathbf{f}_1$ and $\mathbf{f}_2$ are both 24×3 matrices which is a

compact way for representing three process functions:

$$\begin{aligned} \mathbf{f}_1 \boldsymbol{\omega} &= f_{11} * \omega_1 + f_{12} * \omega_2 + f_{13} * \omega_3; \\ \mathbf{f}_2 \mathbf{a} &= f_{21} * a_1 + f_{22} * a_2 + f_{23} * a_3; \end{aligned} \quad (14)$$

To define the basis functions for this system, we follow the conditions of *Theorem 1*: (i) Select β_1 as the measurement function $\mathbf{z}$. (ii) Select the remaining basis functions so that $\frac{\partial \beta}{\partial \mathbf{x}} \cdot \mathbf{f}_i$ are functions of β for all the process functions.¹

The first basis is defined as the measurement function:

$$\beta_1 \triangleq \mathbf{z} = \mathbf{C}_C \mathbf{C}(\mathbf{p}_f - \mathbf{p}) - \mathbf{C}_C \mathbf{p}_C \quad (15)$$

where β_1 is a 3×1 vector representing in a compact form 3 basis functions. The span of β_1 with respect to $\mathbf{x}$ is:

$$\begin{aligned} \frac{\partial \beta_1}{\partial \mathbf{x}} &= \begin{bmatrix} \frac{\partial \beta_1}{\partial \theta} & \frac{\partial \beta_1}{\partial \mathbf{s}} & \frac{\partial \beta_1}{\partial \mathbf{v}} & \frac{\partial \beta_1}{\partial \mathbf{p}} & \frac{\partial \beta_1}{\partial \mathbf{p}_f} & \frac{\partial \beta_1}{\partial \mathbf{b}_a} & \frac{\partial \beta_1}{\partial \mathbf{b}_g} & \frac{\partial \beta_1}{\partial \mathbf{s}_C} & \frac{\partial \beta_1}{\partial \mathbf{p}_C} \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{C}_C [\mathbf{C}(\mathbf{p}_f - \mathbf{p})] \frac{\partial \theta}{\partial \mathbf{s}} & \mathbf{0} & -\mathbf{C}_C \mathbf{C} & \mathbf{C}_C \mathbf{C} & \mathbf{0} & \mathbf{0} & [\beta_1] \frac{\partial \theta_C}{\partial \mathbf{s}_C} & -\mathbf{C}_C \end{bmatrix} \end{aligned}$$

After the span of the first basis function β_1 is obtained, we project it onto all the process functions for finding other basis functions that satisfy condition (A2) of *Theorem 1*. Every time a term in the product can not be expressed by the previously defined basis functions, we incorporate it as a new basis function:

$$\begin{aligned} \frac{\partial \beta_1}{\partial \mathbf{x}} \cdot \mathbf{f}_0 &= -\mathbf{C}_C [\mathbf{C}(\mathbf{p}_f - \mathbf{p})] \mathbf{b}_g - \mathbf{C}_C \mathbf{C} \mathbf{v} \\ &= -[\beta_1 + \mathbf{C}_C \mathbf{p}_C] \mathbf{C}_C \mathbf{b}_g - \mathbf{C}_C \mathbf{C} \mathbf{v} \\ &\triangleq -[\beta_1 + \beta_2] \beta_3 - \beta_4 \\ \frac{\partial \beta_1}{\partial \mathbf{x}} \cdot \mathbf{f}_{1i} &= \mathbf{C}_C [\mathbf{C}(\mathbf{p}_f - \mathbf{p})] \mathbf{e}_i \\ &= [\mathbf{C}_C \mathbf{C}(\mathbf{p}_f - \mathbf{p})] \mathbf{C}_C \mathbf{e}_i \\ &\triangleq [\beta_1 + \beta_2] \beta_{5i} \\ \frac{\partial \beta_1}{\partial \mathbf{x}} \cdot \mathbf{f}_{2i} &= \mathbf{0} \end{aligned}$$

where $i = 1, 2, 3$, $\mathbf{e}_1 = [1 \ 0 \ 0]^T$, $\mathbf{e}_2 = [0 \ 1 \ 0]^T$, $\mathbf{e}_3 = [0 \ 0 \ 1]^T$, $\frac{\partial \theta}{\partial \mathbf{s}} \frac{1}{2} \mathbf{D} = \frac{\partial \theta}{\partial \mathbf{s}} \frac{\partial \mathbf{s}}{\partial \theta} = \mathbf{I}_3$, $\beta_2 = \mathbf{C}_C \mathbf{p}_C$, $\beta_3 = \mathbf{C}_C \mathbf{b}_g$, $\beta_4 = \mathbf{C}_C \mathbf{C} \mathbf{v}$ and $\beta_{5i} = \mathbf{C}_C \mathbf{e}_i$, $i = 1, 2, 3$.

To make the newly defined basis functions satisfy condition (A2), we also project their spans on the process functions. The spans of the basis functions β_2 and β_3 are:

$$\begin{aligned} \frac{\partial \beta_2}{\partial \mathbf{x}} &= \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & [\mathbf{C}_C \mathbf{p}_C] \frac{\partial \theta_C}{\partial \mathbf{s}_C} & \mathbf{C}_C \end{bmatrix} \\ \frac{\partial \beta_3}{\partial \mathbf{x}} &= \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{C}_C & [\mathbf{C}_C \mathbf{b}_g] \frac{\partial \theta_C}{\partial \mathbf{s}_C} & \mathbf{0} \end{bmatrix} \end{aligned}$$

Multiplying either one with any process function $\mathbf{f}_i$ results in zero, thus no new basis functions need to be defined. The span of β_4 , as well as its projections on the process functions are:

$$\frac{\partial \beta_4}{\partial \mathbf{x}} = \begin{bmatrix} \mathbf{C}_C [\mathbf{C} \mathbf{v}] \frac{\partial \theta}{\partial \mathbf{s}} & \mathbf{C}_C \mathbf{C} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & [\mathbf{C}_C \mathbf{C} \mathbf{v}] \frac{\partial \theta_C}{\partial \mathbf{s}_C} & \mathbf{0} \end{bmatrix}$$

¹Note that although the definition of the basis functions is not unique, any basis functions that satisfy the conditions of *Theorem 1* span the same space.

$$\begin{aligned} \frac{\partial \beta_4}{\partial \mathbf{x}} \cdot \mathbf{f}_0 &= -[\mathbf{C}_C \mathbf{C} \mathbf{v}] \mathbf{C}_C \mathbf{b}_g + \mathbf{C}_C \mathbf{C} \mathbf{g} - \mathbf{C}_C \mathbf{b}_a \\ &\triangleq -[\beta_4] \beta_3 + \beta_6 - \beta_7 \end{aligned}$$

$$\begin{aligned} \frac{\partial \beta_4}{\partial \mathbf{x}} \cdot \mathbf{f}_{1i} &= [\mathbf{C}_C \mathbf{C} \mathbf{v}] \mathbf{C}_C \mathbf{e}_i \triangleq [\beta_4] \beta_{5i} \\ \frac{\partial \beta_4}{\partial \mathbf{x}} \cdot \mathbf{f}_{2i} &= \mathbf{C}_C \mathbf{e}_i \triangleq \beta_{5i} \end{aligned}$$

where $\beta_6 = \mathbf{C}_C \mathbf{C} \mathbf{g}$ and $\beta_7 = \mathbf{C}_C \mathbf{b}_a$.

Projecting the span of β_{5i} or β_7 onto any process function results in zero. The span of β_6 , as well as its projections on the process functions are:

$$\begin{aligned} \frac{\partial \beta_6}{\partial \mathbf{x}} &= \begin{bmatrix} \mathbf{C}_C [\mathbf{C} \mathbf{g}] \frac{\partial \theta}{\partial \mathbf{s}} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & [\mathbf{C}_C \mathbf{C} \mathbf{g}] \frac{\partial \theta_C}{\partial \mathbf{s}_C} & \mathbf{0} \end{bmatrix} \\ \frac{\partial \beta_6}{\partial \mathbf{x}} \cdot \mathbf{f}_0 &= -\mathbf{C}_C [\mathbf{C} \mathbf{g}] \mathbf{b}_g = -[\beta_6] \beta_3 \\ \frac{\partial \beta_6}{\partial \mathbf{x}} \cdot \mathbf{f}_{1i} &= \mathbf{C}_C [\mathbf{C} \mathbf{g}] \mathbf{e}_i = [\beta_6] \beta_{5i} \\ \frac{\partial \beta_6}{\partial \mathbf{x}} \cdot \mathbf{f}_{2i} &= \mathbf{0} \end{aligned}$$

which can all be expressed by the previously defined basis functions.

Thus, we have defined all the required bases for the IMU-RGBD camera calibration problem:

$$\begin{aligned} \beta_1 &= \mathbf{C}_C \mathbf{C}(\mathbf{p}_f - \mathbf{p}) - \mathbf{C}_C \mathbf{p}_C & \beta_2 &= \mathbf{C}_C \mathbf{p}_C \\ \beta_3 &= \mathbf{C}_C \mathbf{b}_g & \beta_4 &= \mathbf{C}_C \mathbf{C} \mathbf{v} \\ \beta_{5i} &= \mathbf{C}_C \mathbf{e}_i & \beta_6 &= \mathbf{C}_C \mathbf{C} \mathbf{g} \\ \beta_7 &= \mathbf{C}_C \mathbf{b}_a \end{aligned}$$

which correspond to the landmark position, IMU position, gyroscope bias, IMU velocity, IMU orientation, gravity vector, and accelerometer bias in the RGBD camera's frame of reference. Based on *Theorem 1*, the resulting system in the basis functions is:

$$\begin{aligned} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_{51} \\ \beta_{52} \\ \beta_{53} \\ \beta_6 \\ \beta_7 \end{bmatrix} &= \underbrace{\begin{bmatrix} -[\beta_1 + \beta_2] \beta_3 - \beta_4 \\ \mathbf{0} \\ -[\beta_4] \beta_3 + \beta_6 - \beta_7 \\ \mathbf{0}_{9 \times 1} \\ -[\beta_6] \beta_3 \\ \mathbf{0} \end{bmatrix}}_{\mathbf{g}_0} + \underbrace{\begin{bmatrix} [\beta_1 + \beta_2] \beta_5 \\ \mathbf{0} \\ [\beta_4] \beta_5 \\ \mathbf{0}_{9 \times 3} \\ [\beta_6] \beta_5 \\ \mathbf{0} \end{bmatrix}}_{\mathbf{g}_1} \boldsymbol{\omega} + \underbrace{\begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \beta_5 \\ \mathbf{0}_{9 \times 3} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}}_{\mathbf{g}_2} \mathbf{a} \\ \mathbf{z} &= \beta_1 \end{aligned} \quad (16)$$

where $\beta_5 \triangleq [\beta_{51}^T \ \beta_{52}^T \ \beta_{53}^T]^T$. In the next section, we will show that system (16) is observable by proving its observability matrix Ξ is of full column rank. Therefore, the basis functions β_1 to β_7 correspond to the system's observable modes.

C. Determining the System's Observability Matrix

In this section, we determine the observability matrix $\mathcal{O}$ of system (13) by computing the observability matrix Ξ of system (16) and the derivatives of the basis functions $\mathbf{B}$ separately. First, we prove the matrix Ξ is of full column rank. Then, we find the nullspace of matrix $\mathbf{B}$, which according to *Theorem 1* correspond to the unobservable directions of system (13).

Lemma 2: System (16) is observable.

Proof: See Appendix I.

Since system (16) is observable, we can find the unobservable directions of system (13) from the nullspace of matrix $\mathbf{B}$.

Theorem 3: The IMU-RGBD camera extrinsic calibration system (13) is unobservable, and its unobservable sub-space is spanned by 4 directions corresponding to the IMU-RGBD camera global position and its rotation around the gravity vector in the global frame.

Proof: System (16) satisfies the conditions of *Theorem 1*. Therefore, $\text{null}(\mathcal{O}) = \text{null}(\mathbf{B})$, which spans the unobservable subspace of the original system (13). Stacking the derivatives of the basis functions with respect to the variable $\mathbf{x}$, the matrix $\mathbf{B}$ is defined as:

$$\mathbf{B} = \begin{bmatrix} \mathbf{C}_C[\mathbf{C}(\mathbf{p}_f - \mathbf{p})] \frac{\partial \theta}{\partial \mathbf{s}} & \mathbf{0} & -\mathbf{C}_C \mathbf{C} & \mathbf{C}_C \mathbf{C} & \mathbf{0} & \mathbf{0} & ([\mathbf{C}_C \mathbf{C}(\mathbf{p}_f - \mathbf{p})] - [\mathbf{C}_C \mathbf{p}_C]) \frac{\partial \theta_C}{\partial \mathbf{s}_C} & -\mathbf{C}_C \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & [\mathbf{C}_C \mathbf{p}_C] \frac{\partial \theta_C}{\partial \mathbf{s}_C} & \mathbf{C}_C \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & [\mathbf{C}_C \mathbf{b}_g] \frac{\partial \theta_C}{\partial \mathbf{s}_C} & \mathbf{0} \\ \mathbf{C}_C[\mathbf{C}\mathbf{v}] \frac{\partial \theta}{\partial \mathbf{s}} & \mathbf{C}_C \mathbf{C} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & [\mathbf{C}_C \mathbf{C}\mathbf{v}] \frac{\partial \theta_C}{\partial \mathbf{s}_C} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & [\mathbf{C}_C \mathbf{e}_1] \frac{\partial \theta_C}{\partial \mathbf{s}_C} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & [\mathbf{C}_C \mathbf{e}_2] \frac{\partial \theta_C}{\partial \mathbf{s}_C} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & [\mathbf{C}_C \mathbf{e}_3] \frac{\partial \theta_C}{\partial \mathbf{s}_C} & \mathbf{0} \\ \mathbf{C}_C[\mathbf{C}\mathbf{g}] \frac{\partial \theta}{\partial \mathbf{s}} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & [\mathbf{C}_C \mathbf{C}\mathbf{g}] \frac{\partial \theta_C}{\partial \mathbf{s}_C} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & -\mathbf{C}_C & \mathbf{0} & -[\mathbf{C}_C \mathbf{b}_a] \frac{\partial \theta_C}{\partial \mathbf{s}_C} & \mathbf{0} \end{bmatrix} \quad (17)$$

We proceed by simplifying $\mathbf{B}$ using Gaussian elimination. Note that $\frac{\partial \theta_C}{\partial \mathbf{s}_C}$ is a full rank matrix (its inverse is $\frac{\partial \mathbf{s}_C}{\partial \theta_C}$). Also as proved in Appendix I, the matrix in the seventh column between the fifth and seventh rows has full column rank. Therefore, all the other terms in the seventh column can be eliminated.

Let $\mathbf{A} = [\mathbf{A}_1^T \ \mathbf{A}_2^T \ \mathbf{A}_3^T \ \mathbf{A}_4^T \ \mathbf{A}_5^T \ \mathbf{A}_6^T \ \mathbf{A}_7^T \ \mathbf{A}_8^T]^T$ be the right nullspace of the matrix $\mathbf{B} = \frac{\partial \mathbf{B}}{\partial \mathbf{x}}$. Then, we employ the relation $\mathbf{B}\mathbf{A} = \mathbf{0}$ to determine the elements of $\mathbf{A}$. Specifically, from the second, third, fifth-seventh, ninth rows of the product $\mathbf{B}\mathbf{A}$, we have:

$$\begin{aligned} \mathbf{C}_C \mathbf{A}_8 &= \mathbf{C}_C \mathbf{A}_6 = \mathbf{A}_7 = \mathbf{C}_C \mathbf{A}_5 = \mathbf{0} \\ \Rightarrow \mathbf{A}_5 &= \mathbf{A}_6 = \mathbf{A}_7 = \mathbf{A}_8 = \mathbf{0} \end{aligned} \quad (18)$$

From the eighth row of the product, we have the relation $\mathbf{C}_C[\mathbf{C}\mathbf{g}] \frac{\partial \theta}{\partial \mathbf{s}} \mathbf{A}_1 = \mathbf{0}$, thus $\mathbf{A}_1 = \frac{\partial \mathbf{s}}{\partial \theta} \mathbf{C}\mathbf{g}$ or $\mathbf{A}_1 = \mathbf{0}$.

If $\mathbf{A}_1 = \frac{\partial \mathbf{s}}{\partial \theta} \mathbf{C}\mathbf{g}$, from the fourth row of $\mathbf{B}\mathbf{A}$ we have $\mathbf{A}_2 = -[\mathbf{v}]\mathbf{g}$, and from the first row we have $\mathbf{A}_3 = -[\mathbf{p}]\mathbf{g}$ and $\mathbf{A}_4 = -[\mathbf{p}_f]\mathbf{g}$.

If $\mathbf{A}_1 = \mathbf{0}$, from the fourth row of $\mathbf{B}\mathbf{A}$ we have $\mathbf{A}_2 = \mathbf{0}$, and from the first row $\mathbf{A}_3 = \mathbf{A}_4 = \mathbf{I}$.

In summary, the system's unobservable directions are spanned by the matrix:

$$\mathbf{N} \triangleq \begin{bmatrix} \frac{\partial \mathbf{s}}{\partial \theta} \mathbf{C}\mathbf{g} & \mathbf{0} \\ -[\mathbf{v}]\mathbf{g} & \mathbf{0} \\ -[\mathbf{p}]\mathbf{g} & \mathbf{I} \\ -[\mathbf{p}_f]\mathbf{g} & \mathbf{I} \\ \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} = [\mathbf{N}^g \ \mathbf{N}^p] \quad (19)$$

where the first row is the perturbing terms in $\mathbf{s}$, while the

corresponding term for θ is $\mathbf{C}\mathbf{g}$. Therefore, the unobservable directions are the global position of the IMU-RGBD camera and the point landmark, $\mathbf{N}^p$, and their rotation about the gravity vector, $\mathbf{N}^g$. ■

IV. CALIBRATION ALGORITHM DESCRIPTION

We implement the IMU-RGBD camera extrinsic calibration system (13) with the OC-EKF proposed in [13]. Specifically, in [13] it was shown that due to the linearization errors, the filter gains spurious information along the unobservable directions, which results in smaller uncertainty (that causes the filter to be inconsistent) and larger estimation errors. To address this problem, the authors proposed to modify the Jacobian matrices in both the OC-EKF's propagation and update steps, so as to avoid changing the unobservable directions due to the linearization errors.

We will briefly describe the methodology of the OC-EKF in the following; while the detailed derivations can be found in [13]. In the OC-EKF, the spurious information is removed by enforcing the following observability constraints:

$$\hat{\mathbf{N}}_{k+1} = \hat{\Phi}_k \hat{\mathbf{N}}_k \quad (20)$$

$$\hat{\mathbf{H}}_k \hat{\mathbf{N}}_k = \mathbf{0}, \quad \forall k > 0 \quad (21)$$

where $\hat{\mathbf{N}}_k$ and $\hat{\mathbf{N}}_{k+1}$ are the unobservable directions presented in (19) at time step k and $k+1$, $\hat{\Phi}_k$ is the Jacobian matrix of the system propagation equation (11) at time step k and $\hat{\mathbf{H}}_k$ is the Jacobian matrix of the system measurement equation (12) at time step k .

Due to the linearization error, the Jacobian matrices $\hat{\Phi}'_k$ and $\hat{\mathbf{H}}'_k$ evaluated using the current state estimates in general do not satisfy the observability constraints (20) and (21). However, we can update them using the following process: First, we propagate the state estimate to determine both the $\hat{\mathbf{N}}_k$ and $\hat{\mathbf{N}}_{k+1}$ in closed form. Then, we find a matrix $\hat{\Phi}_k$ that satisfies (20) and has the minimum distance to $\hat{\Phi}'_k$ in the Frobenius norm sense, and use it for the covariance propagation. Similarly, in the update step, we use the current state estimates to compute $\hat{\mathbf{H}}'_k$ and then find the closest matrix $\hat{\mathbf{H}}_k$ in the Frobenius norm sense satisfying (21).

V. SIMULATION AND EXPERIMENTAL RESULTS

We have employed the OC-EKF in both our simulations and experiments, to: (i) Demonstrate the accuracy of our proposed IMU-RGBD camera extrinsic calibration algorithm. (ii) Show that by removing spurious information, the OC-EKF outperforms the standard EKF. (iii) Verify the system's unobservable directions by demonstrating the improvement of the OC-EKF designed using these unobservable directions.

A. Simulation Results

In our simulation, the RGBD camera observes 20 landmarks, which are randomly generated in a plane. The IMU travels in front of this plane with constant local rotational velocity $[2.8648 \ 0 \ 2.8648]^T$ deg/sec and linear acceleration $[0 \ 0.1 \ 0]^T$ m/sec². The initial IMU velocity is

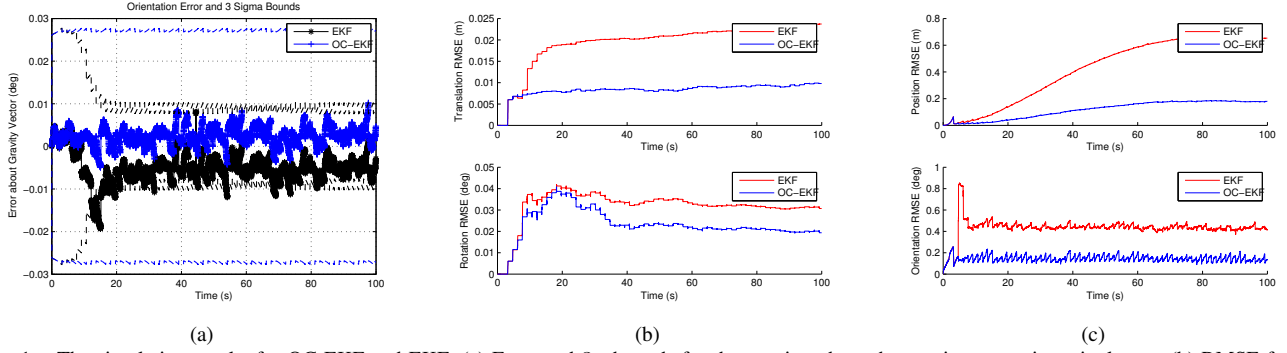


Fig. 1. The simulation results for OC-EKF and EKF: (a) Error and 3σ bounds for the rotation about the gravity vector in a single run. (b) RMSE for the translation and rotation between the two sensors over 20 Monte Carlo trials. (c) RMSE for the robot position and orientation over 20 Monte Carlo trials.

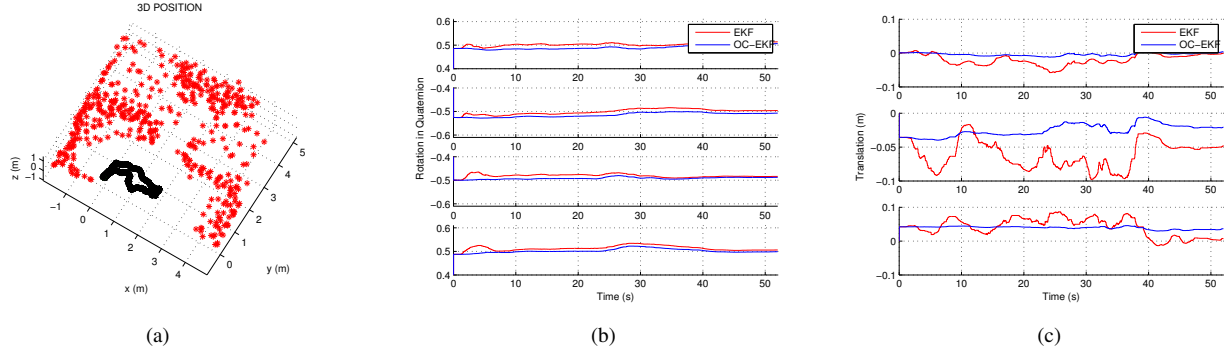


Fig. 2. Experimental results: (a) The platform trajectory and estimated landmarks. (b) The rotation between IMU and kinect in quaternion representation. (c) The translation between the IMU and kinect.

set to $[0.02 \ 0 \ 0.01]^T$ m/sec. The rotation matrix and translation of the IMU in the RGBD camera frame is set to $[-1 \ 0 \ 0; 0 \ 0 \ 1; 0 \ 1 \ 0]$ and $[.1 \ .1 \ .1]^T$ m. The initial standard deviation of the translation and rotation is set to 0.02 m and 2 degrees.

First, we examine the performance of the OC-EKF and EKF in a single run. The estimation error and 3σ bounds of the IMU's orientation around the gravity vector are shown in Fig. 1(a). As evident, the EKF gains spurious information, thus the covariance decreases along with time, while the covariance of the OC-EKF does not. In addition, the EKF becomes inconsistent gradually, while the OC-EKF remains consistent.

In Fig. 1(b) and Fig. 1(c), we present the results of 20 Monte Carlo simulations. The estimation RMSE (root mean square errors) for the IMU position, orientation and the transformation between the two sensors are all smaller for the OC-EKF as compared to the EKF, which verifies the validity of our observability analysis based on which the OC-EKF was designed.

B. Experimental Results

We compare the performance of the OC-EKF with the EKF in practice when performing simultaneous localization, mapping and extrinsic IMU-RGBD camera calibration. In our experimental setup, we utilize a platform comprised of an InterSense NavChip IMU and a kinect, which has both an RGB camera and an infrared (IR) depth-finding camera. The intrinsic parameters of the kinect RGB camera and IR camera, as well as the transformation between the two

cameras, are calibrated offline with the algorithm described in [5].

We train a vocabulary tree [16] with the SIFT features extracted in an indoor environment. During the experiments, the SIFT features of each image are passed through into the vocabulary tree's leaves, and the leaf indexes for each image are recorded. Once a new image is captured, we pass all its SIFT features through the vocabulary tree to its leaves, and find the images with the most similar SIFT features. Then the identities of the SIFT features in the current image are determined by matching them to SIFT features in similar images and employing a geometry consistency test. Since the transformation between the IR camera and the RGB camera is known, the corresponding 3D points captured by the IR camera can also be associated.

In our experiment, the IMU-kinect platform travels in an indoor office along a closed-loop trajectory, which means at the end the platform comes back to its starting location. This information is used to access the estimated robot position accuracy. The experimental results are depicted in Fig. 2. The black line is the platform's trajectory and the red stars denote the estimated landmarks. The whole trajectory is 7.04 m, and the final position error is 7.5 cm for the OC-EKF and 8.1 cm for the EKF. We also plot the estimated translation and rotation between the IMU and kinect in Fig. 2. As evident, the estimated transformation when using the OC-EKF is more stable than that from the EKF.

VI. CONCLUSION

In this paper, we present an observability analysis for the IMU-RGBD camera calibration problem in an unknown

environment using Lie derivatives. Due to the requirement of finding the infinite dimensional observability matrix's nullspace, it is pretty challenge to determine a system's unobservable directions. To address this problem, we decompose the observability matrix into the product of two matrices, Ξ and $\mathbf{B}$, using a nonlinear transformation (basis function) of the original variables. Then we find the system's unobservable directions from the nullspace of the matrix $\mathbf{B}$, which has finite dimension. Moreover, we construct an observable system using the basis functions whose observability matrix is the first matrix Ξ , and show that the basis functions are the IMU-RGBD camera calibration system's observable modes. We present extensive simulation and experimental results to show that the observability-constrained (OC)-EKF designed to adhere to the system's unobservable directions significantly outperforms the EKF.

In our future work, we plan to perform observability analysis of the IMU-camera extrinsic calibration system employing the same methodology and improve its consistency.

APPENDIX I

In this section, we will compute the observability matrix of system (16), and show it is of full column rank; thus system (16) is observable and the basis functions β are the observable modes of the original system (13).

The zeroth order Lie derivative and its span are defined as:

$$\begin{aligned}\mathcal{L}^0 \mathbf{h} &= \beta_1 \\ \nabla \mathcal{L}^0 \mathbf{h} &= \begin{bmatrix} \frac{\partial \mathcal{L}^0 \mathbf{h}}{\partial \beta_1} & \frac{\partial \mathcal{L}^0 \mathbf{h}}{\partial \beta_2} & \frac{\partial \mathcal{L}^0 \mathbf{h}}{\partial \beta_3} & \frac{\partial \mathcal{L}^0 \mathbf{h}}{\partial \beta_4} & \frac{\partial \mathcal{L}^0 \mathbf{h}}{\partial \beta_{5i}} & \frac{\partial \mathcal{L}^0 \mathbf{h}}{\partial \beta_6} & \frac{\partial \mathcal{L}^0 \mathbf{h}}{\partial \beta_7} \end{bmatrix} \\ &= [\mathbf{I} \quad \mathbf{0} \quad \mathbf{0} \quad \mathbf{0} \quad \mathbf{0}_{3 \times 9} \quad \mathbf{0} \quad \mathbf{0}]\end{aligned}$$

which is then used to compute the first-order Lie derivatives and their spans:

$$\begin{aligned}\mathcal{L}_{g_0}^1 \mathbf{h} &= \nabla \mathcal{L}^0 \mathbf{h} \cdot g_0 = [\mathbf{I} \quad \mathbf{0} \quad \mathbf{0} \quad \mathbf{0} \quad \mathbf{0}_{3 \times 9} \quad \mathbf{0} \quad \mathbf{0}] \cdot g_0 \\ &= -[\beta_1 + \beta_2] \beta_3 - \beta_4 \\ \mathcal{L}_{g_{1i}}^1 \mathbf{h} &= \nabla \mathcal{L}^0 \mathbf{h} \cdot g_{1i} = [\mathbf{I} \quad \mathbf{0} \quad \mathbf{0} \quad \mathbf{0} \quad \mathbf{0}_{3 \times 9} \quad \mathbf{0} \quad \mathbf{0}] \cdot g_{1i} \\ &= [\beta_1 + \beta_2] \beta_{5i} \\ \mathcal{L}_{g_{2i}}^1 \mathbf{h} &= \nabla \mathcal{L}^0 \mathbf{h} \cdot g_{2i} = \mathbf{0} \\ \nabla \mathcal{L}_{g_0}^1 \mathbf{h} &= [[\beta_3] \quad [\beta_3] \quad -[\beta_1 + \beta_2] \quad -\mathbf{I} \quad \mathbf{0}_{3 \times 9} \quad \mathbf{0} \quad \mathbf{0}] \\ \nabla \mathcal{L}_{g_{1i}}^1 \mathbf{h} &= [-[\beta_{51}] \quad -[\beta_{51}] \quad \mathbf{0} \quad \mathbf{0} \quad [\beta_1 + \beta_2] \quad \mathbf{0}_{3 \times 6} \quad \mathbf{0} \quad \mathbf{0}] \\ \nabla \mathcal{L}_{g_{12}}^1 \mathbf{h} &= [-[\beta_{52}] \quad -[\beta_{52}] \quad \mathbf{0} \quad \mathbf{0} \quad \mathbf{0} \quad [\beta_1 + \beta_2] \quad \mathbf{0} \quad \mathbf{0}] \\ \nabla \mathcal{L}_{g_{13}}^1 \mathbf{h} &= [-[\beta_{53}] \quad -[\beta_{53}] \quad \mathbf{0} \quad \mathbf{0} \quad \mathbf{0}_{3 \times 6} \quad [\beta_1 + \beta_2] \quad \mathbf{0} \quad \mathbf{0}] \\ \nabla \mathcal{L}_{g_{2i}}^1 \mathbf{h} &= \mathbf{0}\end{aligned}$$

In the observability analysis, the objective is to prove that matrix Ξ is of full column rank. To do so, we only need to consider a subset of the high-order Lie derivatives; the ones that we will use for determining the rank of Ξ , Specifically, the selected second-order Lie derivatives and their gradients are:

$$\begin{aligned}\mathcal{L}_{g_0 g_{2i}}^2 \mathbf{h} &= -\beta_{5i} \\ \mathcal{L}_{g_0 g_0}^2 \mathbf{h} &= -[\beta_3] [\beta_1 + \beta_2] \beta_3 - 2[\beta_3] \beta_4 - \beta_6 + \beta_7 \\ \nabla \mathcal{L}_{g_0 g_{2i}}^2 \mathbf{h} &= [\mathbf{0} \quad \mathbf{0} \quad \mathbf{0} \quad \mathbf{0} \quad -\mathbf{I} \quad \mathbf{0}_{3 \times 6} \quad \mathbf{0} \quad \mathbf{0}] \\ \nabla \mathcal{L}_{g_0 g_{22}}^2 \mathbf{h} &= [\mathbf{0} \quad \mathbf{0} \quad \mathbf{0} \quad \mathbf{0} \quad \mathbf{0} \quad -\mathbf{I} \quad \mathbf{0} \quad \mathbf{0}] \\ \nabla \mathcal{L}_{g_0 g_{23}}^2 \mathbf{h} &= [\mathbf{0} \quad \mathbf{0} \quad \mathbf{0} \quad \mathbf{0} \quad \mathbf{0}_{3 \times 6} \quad -\mathbf{I} \quad \mathbf{0} \quad \mathbf{0}] \\ \nabla \mathcal{L}_{g_0 g_0}^2 \mathbf{h} &= [[\beta_3]^2 \quad [\beta_3]^2 \quad \Pi_3 \quad -2[\beta_3] \quad \mathbf{0} \quad -\mathbf{I} \quad \mathbf{I}]\end{aligned}$$

where Π_3 is the term that will be removed later on through Gaussian elimination, thus we do not show its explicit expression.

The selected third-order Lie derivatives and their gradients are:

$$\begin{aligned}\mathcal{L}_{g_0 g_0 g_{1i}}^3 \mathbf{h} &= [\beta_3]^2 [\beta_1 + \beta_2] \beta_{5i} - 2[\beta_3] [\beta_4] \beta_{5i} - [\beta_6] \beta_{5i} \\ \mathcal{L}_{g_0 g_0 g_{2i}}^3 \mathbf{h} &= -2[\beta_3] \beta_{5i} \\ \nabla \mathcal{L}_{f_0 f_0 f_{1i}}^3 \mathbf{h} &= [-[\beta_3]^2 [\beta_{5i}] \quad -[\beta_3]^2 [\beta_{5i}] \quad \Pi_{3i} \quad 2[\beta_3] [\beta_{5i}] \quad \Pi_{5i} \quad [\beta_{5i}] \quad \mathbf{0}] \\ \nabla \mathcal{L}_{g_0 g_0 g_{21}}^3 \mathbf{h} &= [\mathbf{0} \quad \mathbf{0} \quad 2[\beta_{51}] \quad \mathbf{0} \quad -2[\beta_3] \quad \mathbf{0}_{3 \times 6} \quad \mathbf{0} \quad \mathbf{0}] \\ \nabla \mathcal{L}_{g_0 g_0 g_{22}}^3 \mathbf{h} &= [\mathbf{0} \quad \mathbf{0} \quad 2[\beta_{52}] \quad \mathbf{0} \quad \mathbf{0} \quad -2[\beta_3] \quad \mathbf{0} \quad \mathbf{0}] \\ \nabla \mathcal{L}_{g_0 g_0 g_{23}}^3 \mathbf{h} &= [\mathbf{0} \quad \mathbf{0} \quad 2[\beta_{53}] \quad \mathbf{0} \quad \mathbf{0}_{3 \times 6} \quad -2[\beta_3] \quad \mathbf{0} \quad \mathbf{0}]\end{aligned}$$

where Π_{3i}' and Π_{5i} also do not affect the observability matrix's rank, thus we do not show their specific expressions.

By stacking together (as block rows) the span of the selected Lie derivatives, matrix Ξ is constructed as:

$$\begin{bmatrix} \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ [\beta_3] & [\beta_3] & -[\beta_1 + \beta_2] & -\mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ -[\beta_{51}] & -[\beta_{51}] & \mathbf{0} & \mathbf{0} & [\beta_1 + \beta_2] & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ -[\beta_{52}] & -[\beta_{52}] & \mathbf{0} & \mathbf{0} & \mathbf{0} & [\beta_1 + \beta_2] & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ -[\beta_{53}] & -[\beta_{53}] & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & [\beta_1 + \beta_2] & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & -\mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & -\mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & -\mathbf{I} & \mathbf{0} & \mathbf{0} \\ -[\beta_3]^2 [\beta_{51}] & -[\beta_3]^2 [\beta_{51}] & \Pi_{51}' & 2[\beta_3] [\beta_{51}] & \Pi_{51} & \mathbf{0} & \mathbf{0} & [\beta_{51}] & \mathbf{0} \\ -[\beta_3]^2 [\beta_{52}] & -[\beta_3]^2 [\beta_{52}] & \Pi_{52}' & 2[\beta_3] [\beta_{52}] & \Pi_{52} & \mathbf{0} & \mathbf{0} & [\beta_{52}] & \mathbf{0} \\ -[\beta_3]^2 [\beta_{53}] & -[\beta_3]^2 [\beta_{53}] & \Pi_{53}' & 2[\beta_3] [\beta_{53}] & \Pi_{53} & \mathbf{0} & \mathbf{0} & [\beta_{53}] & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & 2[\beta_{51}] & \mathbf{0} & -2[\beta_3] & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & 2[\beta_{52}] & \mathbf{0} & \mathbf{0} & -2[\beta_3] & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & 2[\beta_{53}] & \mathbf{0} & \mathbf{0} & \mathbf{0} & -2[\beta_3] & \mathbf{0} & \mathbf{0} \end{bmatrix} \quad (22)$$

Since the rank of a matrix does not change after Gaussian elimination, we employ this method to first simplify Ξ and then show that the resulting matrix is of column rank.

We start by providing that $\mathbf{K} \triangleq [[\beta_{51}]^T \quad [\beta_{52}]^T \quad [\beta_{53}]^T]^T$ is of full column rank. To do so, we define a matrix $\mathbf{M}$ as:

$$\mathbf{M} \triangleq \mathbf{C}_c \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \mathbf{C}_c^T \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{1} & \mathbf{0} & \mathbf{0} \end{bmatrix} \mathbf{C}_c^T \begin{bmatrix} \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \mathbf{C}_c^T \quad (23)$$

Recall that $\beta_{5i} = \mathbf{C}_c \mathbf{e}_i$. Multiplying $\mathbf{M}$ from the left by $\mathbf{K}$:

$$\mathbf{MK} = \mathbf{M} \begin{bmatrix} \mathbf{C}_c [\mathbf{e}_1] \\ \mathbf{C}_c [\mathbf{e}_2] \\ \mathbf{C}_c [\mathbf{e}_3] \end{bmatrix} \mathbf{C}_c^T = \mathbf{C}_c \mathbf{I}_3 \mathbf{C}_c^T = \mathbf{I}_3 \quad (24)$$

which means the rank of $\mathbf{MK}$ is 3, thus $\mathbf{K}$ has full column rank.

We now perform block-row Gaussian elimination on matrix Ξ in (22): (i) Using the identity matrix in the (1,1) block element of Ξ , all other terms in its first column can be eliminated. Using the identity matrix in the fifth, sixth

and seventh columns, all other terms in these columns can be eliminated. The resulting matrix is Ξ' :

$$\begin{bmatrix} \mathbf{I} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & [\beta_3] & -[\beta_1 + \beta_2] & -\mathbf{I} & 0 & 0 & 0 & 0 & 0 \\ 0 & -[\beta_{51}] & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -[\beta_{52}] & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -[\beta_{53}] & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\mathbf{I} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\mathbf{I} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\mathbf{I} & 0 & 0 \\ 0 & [\beta_3]^2 & \Pi_3 & -2[\beta_3] & 0 & 0 & 0 & -\mathbf{I} & \mathbf{I} \\ 0 & -[\beta_3]^2[\beta_{51}] & \Pi_{31}'' & 2[\beta_3][\beta_{51}] & 0 & 0 & 0 & [\beta_{51}] & 0 \\ 0 & -[\beta_3]^2[\beta_{52}] & \Pi_{32}'' & 2[\beta_3][\beta_{52}] & 0 & 0 & 0 & [\beta_{52}] & 0 \\ 0 & -[\beta_3]^2[\beta_{53}] & \Pi_{33}'' & 2[\beta_3][\beta_{53}] & 0 & 0 & 0 & [\beta_{53}] & 0 \\ 0 & 0 & 2[\beta_{51}] & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2[\beta_{52}] & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2[\beta_{53}] & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (25)$$

(ii) Between the third and fifth rows of the second column the only left terms are $\mathbf{K}$. By multiplying with $\mathbf{M}$ to these rows, we get the identity matrix. Then, all the other terms in the second column can be eliminated. Similarly, all the rows in the third column except the last three rows can be eliminated. The resulting matrix is Ξ'' :

$$\begin{bmatrix} \mathbf{I} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\mathbf{I} & 0 & 0 & 0 & 0 & 0 \\ 0 & \mathbf{I} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\mathbf{I} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\mathbf{I} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\mathbf{I} & 0 & 0 \\ 0 & 0 & 0 & -2[\beta_3] & 0 & 0 & 0 & -\mathbf{I} & \mathbf{I} \\ 0 & 0 & 0 & 2[\beta_3][\beta_{51}] & 0 & 0 & 0 & [\beta_{51}] & 0 \\ 0 & 0 & 0 & 2[\beta_3][\beta_{52}] & 0 & 0 & 0 & [\beta_{52}] & 0 \\ 0 & 0 & 0 & 2[\beta_3][\beta_{53}] & 0 & 0 & 0 & [\beta_{53}] & 0 \\ 0 & 0 & \mathbf{I} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (26)$$

(iii) The only left term in the second row now is a identity matrix in the fourth column, thus all the other terms in this column can be eliminated:

$$\begin{bmatrix} \mathbf{I} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\mathbf{I} & 0 & 0 & 0 & 0 & 0 \\ 0 & \mathbf{I} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\mathbf{I} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\mathbf{I} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\mathbf{I} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\mathbf{I} & \mathbf{I} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & [\beta_{51}] & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & [\beta_{52}] & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & [\beta_{53}] & 0 \\ 0 & 0 & \mathbf{I} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (27)$$

(iv) With the matrix $\mathbf{K}$ in the eighth column, the identity matrix in the above row can be eliminated. After the

Gaussian elimination, the matrix Ξ becomes:

$$\begin{bmatrix} \mathbf{I} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\mathbf{I} & 0 & 0 & 0 & 0 & 0 \\ 0 & \mathbf{I} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\mathbf{I} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\mathbf{I} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\mathbf{I} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \mathbf{I} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \mathbf{I} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \mathbf{I} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (28)$$

which is obviously a full column rank matrix, thus the system (16) is observable and our defined basis functions are the system's observable modes.

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